Chapter 10

Recent developments in the econometrics of randomized controlled trials☆

Federico A. Bugni[a] **and Yichong Zhang**[b]
[a]*Department of Economics, Northwestern University, Evanston, IL, United States,* [b]*School of Economics, Singapore Management University, Singapore, Singapore*

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Abstract
This paper reviews recent econometric advances in inference for randomized controlled trials (RCTs). Specifically, we focus on a superpopulation framework using a sampling-based approach. We examine two common methods to enhance inference quality in RCTs through baseline covariates: (1) covariate-adaptive randomization during the design stage and (2) regression adjustment during the analysis stage. We also explore how these methods extend to inference in RCTs with imperfect compliance.

1 Introduction

This paper aims to describe recent developments in the econometric analysis of *randomized controlled trials* (RCTs), also known as *randomized experiments*.

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An RCT is an experimental method in which a researcher evaluates the impact of a treatment (i.e., an intervention or a policy) by randomly assigning individuals in the sample to a treatment group or a control group. This random assignment is *controlled* by the researcher, and designed to guarantee that any differences in the distribution of the two groups can be directly attributed to the treatment's effect. As a consequence of this, the RCT allows for the identification and estimation of treatment effect parameters with causal interpretation.

RCTs play a crucial role in all sciences, providing a robust and reliable method for uncovering causal relationships across multiple disciplines. They have a long history in agricultural and biomedical settings. For instance, they are routinely used by the U.S. Food and Drug Administration for the approval of new pharmaceuticals. Within economics, RCTs have been used in several areas of research including public economics (e.g., negative income tax experiments, e.g., see Robins, 1985), health economics (e.g., the RAND health care experiment, e.g., see Manning et al., 1987), experimental economics (e.g., see Kagel and Roth, 1995) and development economics (e.g., see Duflo et al., 2007). In support of this research efforts, there is a vast and very active literature in statistics and econometrics that studies the statistical analysis of RCTs. See, for instance, Fisher (1925, 1935); Neyman (1923); Kempthorne (1952); Rubin (1974, 1978) for some early developments, and Imbens and Rubin (2015); Rosenberger and Lachin (2016) for modern textbook treatments. Finally, we also recommend recent surveys by Athey and Imbens (2017); Bai et al. (2024b).

This paper provides an overview of recent econometric developments regarding inference in RCTs. We consider a "super-population" framework in which the researcher extracts a random sample of data from the population of interest; See Section 1.1 for a further description. We consider a researcher interested in the treatment effect of a binary variable on a particular outcome variable. However, the main ideas in this article extend to RCTs with multiple treatments. For the sake of concreteness, we assume that the researcher is interested in doing inference about the average treatment effect (ATE), which is defined as the expected value of the treatment effect across the entire population of interest. We describe our environment in detail in Section 2.

As part of the data collection to implement the RCT, the researcher obtains a vector of individual characteristics for each RCT participant, known as baseline covariates. The researchers routinely use these baseline covariates in at least two key ways. In the design phase of the RCT, baseline covariates are often used to determine the treatment status of the RCT participants. This is sometimes referred in the literature as covariate adaptive randomization, and it includes treatment assignment practices such as stratification, stratified block randomization, blocking, or paired designs. We analyze RCTs with covariate adaptive randomization in Section 3. Second, in the analysis phase of the RCT, researchers often use baseline covariates to improve on the estimators obtained from the RCT, typically via linear regressions. This is referred to as covariate adjustment and is considered in Section 4. This paper discusses how covariate

adaptive randomization and covariate adjustment affect the statistical properties of the ATE estimators.

Covariate adaptive randomization and covariate adjustment are routinely used in RCTs, and they affect the statistical properties of resulting estimators. Our objective in this paper is to describe the effect that covariate adaptive randomization and covariate adjustment have on the asymptotic properties of the ATE estimators. We approach this topic from both a positive and a normative perspective. We first describe how covariate adaptive randomization and covariate adjustment affect the limiting behavior of the ATE estimators, and then we describe ways in which their use can lead to improvements on their asymptotic properties.

Finally, in practice, it is not uncommon for RCT participants to fail to comply with the treatment assignment provided by the researcher. Individuals assigned to the treatment group may decide not to take the treatment and, similarly, individuals in the control group may find alternative ways of obtaining the treatment. This topic is known as imperfect compliance, and we discuss it in Section 5. We show that imperfect compliance introduces important complications to the identification and inference of the ATE in RCTs. In line with the previous discussion, our analysis of imperfect compliance will allow the RCT to be implemented with covariate adaptive randomization.

1.1 Sampling frameworks and approaches

The literature provides two frameworks for the econometric analysis of RCTs: "finite population" and "super-population", and, relatedly, two inference approaches to inference: "design-based" and "sampling-based". For recent exposition on these, see Abadie et al. (2020, 2023) and Imbens and Rubin (2015, Sections 1.12 and 6.7).

The finite population approach was first considered by Neyman (1923), and is dominant in statistics. In this framework, the data represent a sample of size n extracted without replacement from a finite number N of observations. Of course, we have that $n \leq N$, and $n = N$ occurs when the entire population is sampled. Within this framework, the finite population with N observations represents the target population of interest. This framework is better suited to represent an empirical situation in which the data used by the researcher is a considerable fraction of the target population. Of course, this includes the case in which the entire target population is sampled. An example of this would be the case when the target population is composed of the $N = 50$ U.S. states, and all of them are observed.

The super-population framework is the approach to inference that dominates much of the econometrics practice. In this framework, the data represent an i.i.d. sample of size n from a hypothetical infinite population distribution, referred to as the super-population. (With an infinite super-population, sampling with or without replacement become equivalent.) Within this framework, the

super-population represents the target population of interest. This framework is intended to represent an empirical situation in which the data used by the researcher is a relatively small fraction of the target population. An example of this would be extracting a random sample of households within a large country.

The sampling-based approach and the design-based approach differ in the sources of uncertainty in the RCT. The main distinction between the two lies in the randomness of the potential outcomes of the population of interest. The design-based approach considers these potential outcomes to be fixed (or conditioned upon), while the sampling-based approach allows these to be random. On the other hand, both approaches agree on the randomness of the treatment assignment. The choice of approach is often related to the type of framework under consideration: the super-population framework typically relies on the sampling-based approach, while the finite population framework tends to use the design-based approach.

For concreteness, this paper focuses on a super-population framework with a sampling-based approach to inference.

2 Setup

As explained in Section 1.1, our analysis of RCT is based on a super-population framework in which the researcher selects n individuals at random from a population distribution of interest to participate in the RCT. The individuals are indexed by $i = 1, \ldots, n$.

For simplicity, we consider an RCT that involves a binary treatment, i.e., each RCT participant is either assigned to the treatment or the control group. In principle, most of the results in this paper extend to multi-valued treatments. Let $A_i \in \{0, 1\}$ denote the treatment assignment of individual i, where $A_i = 1$ denotes treatment and $A_i = 0$ denotes control. The basis for inference in an RCT is that the treatment assignment is defined by the researcher to satisfy certain exogeneity assumptions, which will be discussed in the next subsections.

For the outcome variable, we adopt a standard potential outcomes model, also known as the Neyman-Rubin causal model.[1] According to this model, each individual i in the RCT is characterized by a vector of *potential outcomes* denoted by $(Y_i(0), Y_i(1))$, where $Y_i(0)$ denotes the outcome variable if the individual is assigned to control and $Y_i(1)$ denotes the outcome variable if the individual is assigned to treatment.[2] It is important to note that the potential outcomes are never simultaneously observed in reality, as each individual can only be observed under treatment or control. We use Y_i to denote the *observed*

[1] For early references that lay the groundwork for this framework, see Neyman (1923); Rubin (1974, 1978, 1980). See also Holland (1986); Rubin (2005) and Imbens and Rubin (2015, Chapter 1) for later expositions.

[2] The notation imposes SUTVA, i.e., Stable Unit Treatment Value Assumption, which embodies two conditions: (i) potential outcomes are not dependent on other individual's treatment assignment and (ii) there is a unique form or version of the treatment. See Rubin (1980) and Imbens and Rubin (2015, Chapter 1.6).

outcome for individual i, which is determined by the potential outcomes and the treatment assignment according to

$$Y_i = Y_i(1)A_i + Y_i(1)(1 - A_i).$$

Finally, we use X_i to denote the vector of baseline covariates of individual i.

The treatment effect of individual i is defined as the difference between the potential outcome with and without treatment:

$$Y_i(1) - Y_i(0). \tag{1}$$

By definition, this represents the causal effect of the treatment on the outcome. We note that the treatment effect of interest is a function of both potential outcomes which are never simultaneously observed. This is often referred to as the fundamental problem of causal inference (e.g., Holland, 1986, P. 947).

In this context, parameters of interest are aggregate measures of the distribution of treatment effect in (1), such as the mean, the median, or its overall distribution. By far, the most common parameter of interest is ATE, which is given by

$$\theta_{\text{ATE}} = E[Y_i(1) - Y_i(0)]. \tag{2}$$

By definition, θ_{ATE} represents the average treatment effect in the population of interest when the treatment is mandated for everyone in the population.

Under our super-population approach to inference, the n RCT participants are a random sample of the population of interest P. As a corollary, $\{(Y_i(0), Y_i(1), X_i)\}_{i=1}^n$ represents an i.i.d. sequence of random variables. On the other hand, the distribution of the observed data $\{(Y_i, X_i, A_i)\}_{i=1}^n$ depends on the properties of the treatment assignment $\{A_i\}_{i=1}^n$, determined by the researcher.

Example 2.1 (Running example). For illustration, we consider a hypothetical RCT inspired by the National Job Corps Study, which evaluated the effectiveness of the Job Corps program—a vocational training initiative for disadvantaged U.S. individuals aged 16 to 24. Participants were randomly assigned to either a treatment group, which received the training, or a control group, which did not. The study measured outcomes such as employment status and earnings for up to two years after the intervention, showing that Job Corps had a positive impact on participants' economic and social well-being (Schochet et al., 2001). In our hypothetical RCT, a sample of n individuals is drawn i.i.d. from the population of U.S. individuals aged 16 to 24, with a subset selected for the training program. For each individual $i = 1, \ldots, n$, $Y_i(0)$ represents employment status after two years without training, $Y_i(1)$ with training, $A_i \in 0, 1$ indicates whether they received the training, Y_i denotes the observed employment status, and X_i includes pre-RCT characteristics such as age, gender, race, income, location, and education. □

2.1 Results without using baseline covariates

The remainder of the paper considers various ways in which baseline covariates are routinely used in the design and analysis of RCTs. Our results will show that using baseline covariates appropriately can improve the quality of the inference in an RCT. For the sake of having a benchmark, we now consider the asymptotic behavior of the standard ATE estimator when baseline covariate information is unused.

We now consider a treatment assignment rule that is statistically independent of covariate information. The simplest possible treatment assignment rule in this regard is *simple random assignment* (SRA), also known as *Bernoulli trials*; see Imbens and Rubin (2015, Section 4.3). Hereafter, let $\pi \in (0, 1)$ denote the proportion of the sample treated units desired by the researcher.

Example 2.2 (Simple random assignment (SRA)). SRA refers to a treatment assignment where $\{A_i\}_{i=1}^n$ is sampled independently from the RCT information, and i.i.d. according to a Bernoulli distribution with parameter $\pi \in (0, 1)$. That is,

$$P\left(\{A_i\}_{i=1}^n = \{a_i\}_{i=1}^n \mid \{(X_i, Y_i(0), Y_i(1))\}_{i=1}^n\right) = \prod_{i=1}^n \pi^{a_i} (1-\pi)^{1-a_i} \tag{3}$$

for any $\{a_i\}_{i=1}^n \in \{0, 1\}^n$. □

We also consider an ATE estimator that makes no use of baseline covariates. A natural candidate for this respect is as follows:

$$\hat{\theta}_{\text{OLS}} = \frac{\widehat{\text{Cov}}(A, Y)}{\widehat{\text{Var}}(A)} = \sum_{i=1}^n \frac{Y_i A_i}{n_1} - \sum_{i=1}^n \frac{Y_i (1 - A_i)}{n - n_1} \tag{4}$$

where $n_1 = \sum_{i=1}^n A_i$ denotes total number of treated units. It is easy to see that $\hat{\theta}_{\text{OLS}}$ is the slope estimator in the linear regression:

$$Y_i = \alpha + \beta A_i + \varepsilon_i \quad \text{for } i = 1, \ldots, n. \tag{5}$$

The second equality in (4) shows that $\hat{\theta}_{\text{OLS}}$ is the difference of sample means between the treated and untreated groups.

Under SRA and finite second-moments for the potential outcomes, $\hat{\theta}_{\text{OLS}}$ is an unbiased, consistent, and asymptotically normal estimator of the ATE. In particular, a standard application of the CLT gives

$$\sqrt{n}(\hat{\theta}_{\text{OLS}} - \theta_{\text{ATE}}) \xrightarrow{d} N(0, \Sigma_{\text{OLS}}) \tag{6}$$

with

$$\Sigma_{\text{OLS}} = \frac{\text{Var}[Y(1)]}{\pi} + \frac{\text{Var}[Y(0)]}{1 - \pi}. \tag{7}$$

Furthermore, in the absence of baseline covariate information, Hirano et al. (2003, Theorem 1) implies that $\hat{\theta}_{\text{OLS}}$ is asymptotically efficient. In addition, it is easy to construct an estimator $\hat{\Sigma}_{\text{OLS}}$ that can consistently estimate Σ_{OLS}. By combining (7) and $\hat{\Sigma}_{\text{OLS}}$, we can implement asymptotically-valid inference for $H_0 : \theta_{\text{ATE}} = \bar{\theta}$ vs. $H_1 : \theta_{\text{ATE}} \neq \bar{\theta}$ for any $\bar{\theta} \in \mathbb{R}$. In particular, for any significance level $\alpha \in (0, 1)$, we propose to reject $H_0 : \theta_0 = \bar{\theta}$ according to the following rejection rule:

$$\phi_n = I\left[\left|\frac{\sqrt{n}(\hat{\theta}_{\text{OLS}} - \bar{\theta})}{\hat{\Sigma}_{\text{OLS}}^{1/2}}\right| > z_{1-\alpha/2}\right], \tag{8}$$

where $z_{1-\alpha/2}$ denotes the $(1 - \alpha/2)$-quantile of $N(0, 1)$.

It is relevant to mention that these findings would repeat if we replaced SRA in (3) with another treatment assignment mechanism known as *completely random assignment* (CRA), defined next.

Example 2.3 (Completely random assignment). For a given $\pi \in (0, 1)$, CRA refers to a treatment assignment rule in which exactly $n_1 = \lfloor \pi n \rfloor$ individuals are assigned to treatment and the remaining $n_0 = n - n_1$ individuals are assigned to control, and where all such possible assignments are equally likely. That is,

$$P\left(\{A_i\}_{i=1}^n = \{a_i\}_{i=1}^n \mid \{(X_i, Y_i(0), Y_i(1))\}_{i=1}^n\right) = \binom{n}{n_1}^{-1} 1\left[\sum_{i=1}^n a_i = n_1\right],$$

for any $\{a_i\}_{i=1}^n \in \{0, 1\}^n$. □

SRA and CRA share the fact that $\{A_i\}_{i=1}^n$ is independent of the baseline covariate information $\{X_i\}_{i=1}^n$. However, it is not hard to see that SRA and CRA are not equivalent. For example, $n_1 = \sum_{i=1}^n A_i$ is a Binomial random variable with parameter (n, π) in SRA and the constant $\lfloor \pi n \rfloor$ in CRA. Despite these differences, $\hat{\theta}_{\text{OLS}}$ is also an unbiased, consistent, and asymptotically normal estimator of the ATE θ that satisfies (6).

This section has characterized the statistical properties of a specific estimator, $\hat{\theta}_{\text{OLS}}$, in the context of RCT that is implemented with simple forms of treatment assignment (SRA or CRA). Both estimator and treatment assignment are "simple" in the sense that they make no use of baseline covariate information. In the remainder, we will show that exploiting this information can produce efficiency gains. In particular, the next sections will describe estimators of θ_{ATE} that are also consistent and asymptotically normal, but with a smaller asymptotic variance than Σ_{OLS}. By standard arguments, these efficiency gains translate directly into more powerful hypothesis tests.

The remainder of this paper will consider the two conceptually distinct ways of using baseline covariate information to improve inference in an RCT. On the one hand, one can use covariate information in the design stage of the RCT by assigning treatments according to covariate adaptive randomization, as discussed in Section 3. On the other hand, one can use covariate information in the analysis stage of the RCT by using covariate adjustment; as described in Section 4. Both of these are routinely used in RCTs in practice, and our results show how these affect the statistical properties of the ATE estimators.

3 Covariate-adaptive randomization

This section introduces a framework for inference on the ATE in the RCT known as covariate-adaptive randomization (CAR). This refers to a treatment assignment scheme that first uses baseline covariates to partition observations into several *strata*, and then assigns treatments to achieve "balance" within each stratum. The term balance here means that the researcher assigns treatments in the RCT so that the proportion of treated units in each stratum is close to the desired target treatment probability π.

The CAR framework that we describe is arguably very general. In particular, it is flexible enough to include several treatment assignment methods that are routinely used to implement RCTs. In economics, the most prominent example of the CAR is *stratification* or *stratified block randomization* (SBR). A notable feature of SBR and many other examples of treatment assignments allowed by the CAR framework is that observed treatments and, consequently, the data, are not i.i.d., which complicates the econometric analysis.

The CAR literature considers two asymptotic frameworks for their strata. On one hand, we have the case in which the researcher constructs a finite number of strata, each composed of a large (i.e., divergent) number of individuals. We refer to this as the "big strata" framework. On the other hand, we have the case in which the researcher constructs strata with a finite number of individuals, and so the number of strata then diverges with the sample size. We refer to this as the "small strata" framework. Both cases are empirically relevant and require slightly different notations, assumptions, and asymptotic arguments, so we consider them in separate sections.

3.1 Big strata

This section is based on the framework proposed by Bugni et al. (2018, 2019). Given the baseline covariate data $\{X_i\}_{i=1}^n$ with support $\mathcal{X}$, the researcher defines strata according to a pre-specified function $S : \mathcal{X} \to \mathcal{S} = \{1, \ldots, |\mathcal{S}|\}$, where $|\mathcal{S}| \in \mathbb{N}$. According to this function, individual i belongs to strata $S_i = S(X_i)$. We avoid having irrelevant strata by assuming that $p(s) \equiv P(S_i = s) > 0$ for all $s \in \mathcal{S}$. We refer to this setup as the "big strata" framework because there are a finite number of strata, and each stratum has a positive probability mass.

Example 3.1 (Running example, continued). In the context of the hypothetical RCT considered in Example 2.1, consider $X_i = (X_{i,1}, X_{i,2}) \in \mathcal{X} = \{0, 1\} \times \{16, 17, \ldots, 24\}$, where $X_{i,1} \in \{0, 1\}$ denotes individual i's gender (say, $X_{i,1} = 0$ is male and $X_{i,1} = 1$ is female) and $X_{i,2} \in \{16, 17, \ldots, 24\}$ denotes individual i's age. The researcher could then stratify individuals into $|\mathcal{S}| = 18$ strata as follows: any female aged $j = 16, \ldots, 24$ would be assigned to strata $s = j - 15$ and any male aged $j = 16, \ldots, 24$ would be assigned to strata $s = 8 + (j - 15) = j - 7$. Then, $S : \mathcal{X} \to \mathcal{S} = \{1, \ldots, 18\}$ is given by

$$S(x_1, x_2) = (x_2 - 15) \times 1[x_1 = 0] + (x_2 - 7) \times 1[x_1 = 1].$$

This example only considers two discrete covariates for simplicity. Of course, including additional covariates would increase the number of strata and adding continuous covariates (e.g., income) would require discretization in order to stratify them. □

The CAR framework then allows the researcher to assign treatment within each stratum to achieve a proportion $\pi \in (0, 1)$ of treated units with a desired level of balance. To define this formally, we require some additional notation. For each stratum $s \in \mathcal{S}$, let $n_1(s)$ be the number of treated units in strata s and let $n(s)$ be the number of units in strata s, and so $n_1(s)/n(s)$ denotes the sample frequency of treated units in strata s. Thus, we can measure the amount of treatment assignment imbalance in stratum s relative to the target proportion π using the following measure:

$$\sqrt{n}\left(n_1(s)/n(s) - \pi\right).$$

To describe a general class of CAR mechanisms, we impose the following assumption:

Assumption 3.1 (CAR mechanisms with big strata). The treatment assignment $\{A_i\}_{i=1}^n$ satisfies the following:

(a) Conditional on $\{S_i\}_{i=1}^n$, $\{(X_i, Y_i(1), Y_i(0))\}_{i=1}^n$ are $\{A_i\}_{i=1}^n$ independent.

(b) Conditional on $\{S_i\}_{i=1}^n$, $\left\{\sqrt{n}\left(n_1(s)/n(s) - \pi\right) : s \in \mathcal{S}\right\} \xrightarrow{d} N(0, diag\{\tau(s)/p(s) : s \in \mathcal{S}\})$ (with probability approaching one), where $\tau(s) \in [0, \pi(1-\pi)]$ for all $s \in \mathcal{S}$.

Assumption 3.1(a) requires that the treatment assignment vector $\{A_i\}_{i=1}^n$ is a function of the strata vector $\{S_i\}_{i=1}^n$ and a randomization device that is exogenous to the sample. Assumption 3.1(b) that the scaled vector of measures of imbalances $\left\{\sqrt{n}\left(n_1(s)/n(s) - \pi\right) : s \in \mathcal{S}\right\}$ converges in distribution to a normal random variable with mean zero and a diagonal variance-covariance matrix. As we will argue next, this limiting result occurs naturally in CAR mechanisms used in practice, both in clinical trials and development economics. The limiting distribution formalizes the requirement that the randomization scheme assigns treatment status within each stratum so that the fraction of units being assigned to treatment within each stratum has a well-behaved distribution centered around the target proportion π as the sample size tends to infinity. The key parameters in this distribution are $\{\tau(s) : s \in \mathcal{S}\}$, which characterize the degree of imbalance allowed by the CAR mechanism. In particular, if $\tau(s) = 0$ for all $s \in \mathcal{S}$, then $\sqrt{n}\left(n_1(s)/n(s) - \pi\right)$ is very concentrated around zero, implying that $n_1(s)/n(s)$ is very concentrated around π. Any CAR mechanism that has this property is said to be a CAR with "strong balance". On the other hand, $\tau(s) > 0$ for some $s \in \mathcal{S}$, implies that $n_1(s)/n(s)$ is not as concentrated around π.

As explained in Bugni et al. (2018), there are several treatment assignment mechanisms that satisfy Assumption 3.1. First, it is not hard to verify simple random sampling (SRA) (Example 2.2) and the completely randomized assignment (Example 2.3) satisfy Assumption 3.1 with $\tau(s) = \pi(1-\pi)$ for all $s \in \mathcal{S}$. That is, our current framework includes both of the simple treatment assignments that do not make use of the covariates. Since neither of them exploits these covariates, it is not surprising that they produce an imbalance measure with the maximum possible value of the limiting variance in Assumption 3.1(b). More interestingly, we now describe a few examples of treatment assignments that satisfy Assumption 3.1 and satisfy the "strong balance" property: stratified block randomization, minimization methods, and Efron's biased coin design.

Example 3.2 (Stratified block randomization). Stratified block randomization (SBR) is also known as permuted blocks within strata or blocking. By definition SBR implements a completely randomized assignment (CRA) within each stratum. That is, for each stratum s with sample size $n(s)$, SBR assigns exactly $n_1(s) = \lfloor \pi n(s) \rfloor$ individuals are assigned to treatment and the remaining $n_0(s) = n(s) - n_1(s)$ individuals are assigned to control, and where all such possible assignments are equally likely. That is, for any $\{a_i\}_{i=1}^n \in \{0,1\}^n$,

$$P\left(\{A_i\}_{i=1}^n = \{a_i\}_{i=1}^n \mid \{(X_i, Y_i(0), Y_i(1))\}_{i=1}^n\right)$$
$$= \prod_{s \in \mathcal{S}} \begin{pmatrix} n(s) \\ n_1(s) \end{pmatrix}^{-1} 1\left[\sum_{i=1}^n a_i = n_1(s)\right].$$

It is not difficult to see that SBR satisfies $|n_1(s) - n(s)\pi| \leq 1$ for all $s \in \mathcal{S}$. From here, we can easily conclude that SBR satisfies Assumption 3.1 with $\tau(s) = 0$ for all $s \in \mathcal{S}$. □

Example 3.3 (Minimization methods). Another popular class of treatment assignment mechanisms is the so-called minimization methods. These were originally proposed by Pocock and Simon (1975), and extended and further investigated by Hu and Hu (2012). In these methods, the treatment is assigned recursively for individual $i = 1, \ldots, n$ according to

$$P(A_i = 1 \mid \{(X_i, Y_i(0), Y_i(1))\}_{i=1}^n, \{A_j\}_{j=1}^{i-1})$$
$$= \begin{cases} \pi & \text{if } Imb_i(\{S_j\}_{j=1}^i, \{A_j\}_{j=1}^{i-1}) = 0 \\ \lambda & \text{if } Imb_i(\{S_j\}_{j=1}^i, \{A_j\}_{j=1}^{i-1}) < 0 \\ 1-\lambda & \text{if } Imb_i(\{S_j\}_{j=1}^i, \{A_j\}_{j=1}^{i-1}) > 0 \end{cases}, \tag{9}$$

where $\{A_j\}_{j=1}^0 = \emptyset$, $\lambda \in (\pi, 1]$, and $Imb_i(\{S_j\}_{j=1}^i, \{A_j\}_{j=1}^{i-1})$ is a weighted measure of imbalance relative to the desired treatment assignment distribution. In particular, Hu and Hu (2012, Page 1797) specify $Imb_i(\{S_j\}_{j=1}^i, \{A_j\}_{j=1}^{i-1})$ as the

weighted sum of three sources of imbalance: (i) overall imbalance, with associated weight w_o, (ii) imbalance within the marginal distribution of the L relevant covariates, each with associated weights $(w_{m,\ell} : \ell = 1, \ldots, L)$, and (iii) within strata imbalance, with associated weight w_s. As explained in Hu and Hu (2012, Remark 2.1), their minimization method includes Pocock and Simon (1975) as a special case when we set $w_o = w_s = 0$. Hu and Hu (2012, Theorem 3.2) provide conditions on $(w_o, w_s, (w_{m,\ell} : \ell = 1, \ldots, L)')$ under which the strata-specific imbalance is a positive recurrent Markov chain, implying that they converge to zero at a fast rate. Thus, under these conditions, this method satisfies Assumption 3.1 with $\tau(s) = 0$ for all $s \in \mathcal{S}$. □

Example 3.4 (Efron's balanced coin design). Efron's balanced-coin design is a random assignment methodology originally proposed by Efron (1971) with the aim of improving balance in finite samples. This method is a special case of Example 3.3 with

$$Imb_i(S^{(i)}, A^{(i-1)}) = \sum_{k=1}^{i-1} (A_k - \pi)\, I\,[S_k = S_i]\,.$$

Bugni et al. (2018, Lemma B.11) shows that this treatment assignment method satisfies Assumption 3.1 with $\tau(s) = 0$ for all $s \in \mathcal{S}$. □

Bugni et al. (2018) derives the asymptotic distribution of regression-based estimands under our super-population framework and the CAR framework described in Assumption 3.1. It is relevant to note that these results do not follow standard asymptotic arguments as the observed data are not necessarily i.i.d. In any case, the results in Bugni et al. (2018) imply that $\hat{\theta}_{\text{OLS}}$ is a consistent and asymptotically normal estimator of the ATE. In particular, the proof of Bugni et al. (2018, Theorem 4.1) implies that

$$\sqrt{n}(\hat{\theta}_{\text{OLS}} - \theta_{\text{ATE}}) \xrightarrow{d} N\left(0, \Sigma_{\text{BS}}(\pi)\right), \tag{10}$$

where $\Sigma_{\text{BS}}(\pi)$ is the asymptotic variance for "big strata" CAR methods, given by

$$\begin{aligned}\Sigma_{\text{BS}}(\pi) &= \tfrac{\text{Var}[Y(1)]}{\pi} + \tfrac{\text{Var}[Y(0)]}{1-\pi} \\ &\quad - E\left[\left(\pi\,(1-\pi) - \tau\,(S_i)\right)\left[\tfrac{E[Y_i(1)|S_i]-E[Y_i(1)]}{\pi} + \tfrac{E[Y_i(0)|S_i]-E[Y_i(0)]}{1-\pi}\right]^2\right] \\ &= \Sigma_{\text{OLS}} - \Delta_{\text{CAR}},\end{aligned} \tag{11}$$

with Σ_{OLS} as in (7) and

$$\begin{aligned}\Delta_{\text{CAR}} &= E\left[\left(\pi\,(1-\pi) - \tau\,(S_i)\right)\left[\tfrac{E[Y_i(1)|S_i]-E[Y_i(1)]}{\pi} + \tfrac{E[Y_i(0)|S_i]-E[Y_i(0)]}{1-\pi}\right]^2\right] \\ &\geq 0.\end{aligned}$$

The asymptotic variance in (10) is equal to that in (6) minus the non-negative term Δ_{CAR}. This term is determined by the CAR design through the degree of imbalance $\{\tau(s) : s \in \mathcal{S}\}$. Along this dimension, the most efficient CAR mechanism satisfies strong balance, i.e., $\tau(s) = 0$ for all $s \in \mathcal{S}$. In that case with "big strata'' and "strong balance", the asymptotic variance is given by

$$\Sigma_{\text{BS,SB}}(\pi) = \frac{\text{Var}[Y(1)]}{\pi} + \frac{\text{Var}[Y(0)]}{1-\pi} - \pi(1-\pi) E\left[\left[\frac{E[Y_i(1)|S_i]-E[Y_i(1)]}{\pi} + \frac{E[Y_i(0)|S_i]-E[Y_i(0)]}{1-\pi}\right]^2\right]. \tag{12}$$

Furthermore, under the assumption that the strata exhaust all the information in the baseline covariates (i.e., there is a one-to-one mapping between the baseline covariates and the strata), results in Armstrong (2022) and Rafi (2023) imply that a CAR mechanism that satisfies strong balance achieves efficiency, i.e., $\Sigma_{\text{BS,SB}}(\pi)$ is the semiparametric efficiency bound.

Bugni et al. (2018) provides a consistent estimator of the asymptotic variance in (10). We denote this estimator by $\hat{\Sigma}_{\text{BS}}(\pi)$; see Bugni et al. (2018, Eq. (20)-(22)) for details. By combining this estimator with (10), we can implement asymptotically valid inference for $H_0 : \theta_0 = \bar{\theta}$ vs. $H_1 : \theta_0 \neq \bar{\theta}$ for any $\bar{\theta} \in \mathbb{R}$. Specifically, for any significance level $\alpha \in (0, 1)$, we propose to reject $H_0 : \theta_0 = \bar{\theta}$ according to the following rejection rule:

$$\phi_n = I\left[\left|\frac{\sqrt{n}(\hat{\theta}_{\text{OLS}} - \bar{\theta})}{\hat{\Sigma}_{\text{BS}}(\pi)^{1/2}}\right| > z_{1-\alpha/2}\right], \tag{13}$$

where $z_{1-\alpha/2}$ denotes the $(1-\alpha/2)$-quantile of $N(0, 1)$.

The results thus far allow us to draw a general conclusion about the effect of using CAR methods to implement an RCT. CAR methods generally lead to weakly more efficient ATE inference relative to simple treatment assignments that do not use covariates (e.g., SRA or CRA). By enforcing balance in the treatment assignment within each stratum, the CAR method can reduce the variability in the asymptotic distribution. In the case of a strongly balancing CAR method, the effect on the asymptotic distribution is the same as if we had conditioned on the strata information. As we will discuss in later sections, one can achieve a similar efficiency result in the analysis stage via regression adjustment.

We have seen that a strongly balancing CAR method has the effect of conditioning on the strata information. This suggests that one can improve the efficiency of an RCT by making strata smaller during the design stage. In other words, if we have two RCTs implemented with a strongly balancing CAR method, and the strata in the first RCT form a finer partition of the baseline information than the strata in the second RCT, then the ATE estimator $\hat{\theta}_{\text{OLS}}$ in the first RCT will be (weakly) more efficient than the corresponding estimator in the second RCT. This result was formally established in RCTs with

imperfect compliance by Bugni and Gao (2023). This logic naturally leads to the recommendation of RCTs with increasingly finer strata. If the baseline covariates have finite support, then this leads to the recommendation that strata in the RCT should map one-to-one with this support. On the other hand, if the baseline covariates have infinitely many points of support, this recommendation would be out of the scope of the current asymptotic framework with large strata (which presumes a finite number of strata with a divergent number of RCT participants in each stratum). To implement this recommendation at face value, one must consider an asymptotic framework with small strata, which naturally segues into the next section.

3.2 Small strata

This section considers causal inference under small strata randomization, where strata sizes are fixed, and the number of strata is proportional to the sample size. We focus on a particular treatment assignment mechanism, known as the *matched tuple design* (MTD). This is a CAR method with small strata defined by two constants: $k \in \mathbb{N}$ and $v \in \{1, \ldots, k-1\}$. In the MTD, the n RCT participants are grouped into sets of size k, referred to as *tuples*. For each tuple, v individuals are randomly assigned to the treatment group, while the remaining $k - v$ individuals are assigned to the control group. In the notation used in Section 3.1, the MTD produces $|\mathcal{S}_n| = \lfloor n/k \rfloor$ strata, which diverges as the sample size grows.[3] Each stratum contains $n(s) = k$ observations with $n_1(s) = v$ treated observations, both of which are fixed as the sample size grows. By design, the MTD results in a target proportion of $\pi = v/k$. Moreover, MTD is a CAR that achieves "strong balance", as the proportion of treated units in each stratum is exactly equal to $\pi = v/k$.

Following Bai (2022) and Bai et al. (2023a), this section will focus on the leading case of the MTD, known as the *matched pair design* (MPD). The MPD is the MTD with $k = 2$ and, necessarily, $v = 1$ (recall that $v \in \{1, \ldots, k-1\} = \{1\}$ when $k = 2$). By definition, the MPD produces $|\mathcal{S}_n| = \lfloor n/2 \rfloor$ strata, and each stratum has exactly two observations, with one unit assigned to treatment and the other to control. By construction, the matched pair design results in a target proportion of $\pi = 1/2$.

Example 3.5 (Running example, continued). In the context of the hypothetical RCT considered in Example 2.1, consider $X_i \in \mathcal{X} = \{0, 1\} \times \mathbb{R}_+$, where $X_{i,1} \in \{0, 1\}$ denotes individual i's gender (say, $X_{i,1} = 0$ is male and $X_{i,1} = 1$ is female) and $X_{i,2} \in \mathbb{R}_+$ denotes individual i's income. The researcher could then divide the sample into the two genders. The researcher would then sort the male subsample by income and form strata by grouping them into pairs of consecutive incomes. The first pair of males with the smallest income would form the first

[3] Since n is not necessarily divisible by k, the MTD leaves $n - k\lfloor n/k \rfloor$ leftover units, which are negligible for our asymptotic analysis.

stratum, the next pair would form the second stratum, and so on. Once all the males are sorted into pairs, the researcher will repeat the process with the female subsample. Once this procedure is completed, the entire sample is divided into pairs, possibly up to a small number of observations, which can be discarded. By construction, each stratum comprises a pair of individuals of the same gender with "close" income levels. The treatment assignment in this example would become more involved if the researcher decided to stratify with additional continuous and discrete covariates. In this respect, Cytrynbaum (2023) provides an algorithm to form strata according to general forms of treatment assignments, including those with small strata. □

Throughout this section, we focus on causal inference under MTDs based on the works of Bai (2022), Bai et al. (2023a), and Cytrynbaum (2023). Following this work, we assume that the treatment assignment rule satisfies the following assumption.

Assumption 3.2 (CAR mechanism with small strata: MPD). The treatment assignment $\{A_i\}_{i=1}^n$ satisfies the following:

(a) Conditional on $\{X_i\}_{i=1}^n$, $\{A_i\}_{i=1}^n$ and $\{(Y_i(1), Y_i(0))\}_{i=1}^n$ are independent.

(b) The RCT participants are organized into $|\mathcal{S}_n| = [n/2]$ pairs. For units in the pair satisfy

$$\frac{1}{n}\sum_{s=1}^{[n/2]} \|X_{s,2} - X_{s,1}\|_2^2 \xrightarrow{P} 0,$$

where $(X_{s,1}, X_{s,2})$ are the observations of X_i in pair $s = 1, \ldots, |\mathcal{S}_n|$. Let $(A_{s,1}, A_{s,2})$ denote the assignments of the two individuals in pair $s = 1, \ldots, |\mathcal{S}_n|$.

(c) Conditional on $\{X_i\}_{i=1}^n$, $\left\{\left(A_{s,1}, A_{s,2}\right)\right\}_{s=1}^{[n/2]}$ are i.i.d. and

$$P\left(\left(A_{s,1}, A_{s,2}\right) = (0,1)\right) = P\left(\left(A_{s,1}, A_{s,2}\right) = (1,0)\right) = 1/2.$$

Under Assumption 3.2 and other regularity conditions, Bai (2022) and Bai et al. (2023a) derive the asymptotic distribution of the OLS estimator in (4). In particular, they show that

$$\sqrt{n}(\hat{\theta}_{\text{OLS}} - \theta_{\text{ATE}}) \xrightarrow{d} N(0, \Sigma_{\text{SS}}\left(\tfrac{1}{2}\right)), \tag{14}$$

where Σ_{SS} is the asymptotic variance of "small strata" CAR methods with $\pi = 1/2$, given by

$$\Sigma_{\text{SS}}\left(\tfrac{1}{2}\right) = \text{Var}[E[Y_i(1) - Y_i(0)|X_i]] + 2E[\text{Var}[Y_i(1)|X_i] + \text{Var}[Y_i(0)|X_i]]. \tag{15}$$

It is illustrative to compare the asymptotic variances of "small strata" and "big strata" CAR methods. To this end, let $\Sigma_{\mathrm{BS}}\left(\frac{1}{2}\right)$ denote Σ_{BS} in (11) specialized to the case of $\pi = 1/2$, and let $\Sigma_{\mathrm{BS,SB}}\left(\frac{1}{2}\right)$ denote Σ_{BS} in (11) specialized to the case of $\pi = 1/2$ and "strong balance" (i.e., $\tau(S_i) = 0$). Then,

$$\begin{aligned}\Sigma_{\mathrm{BS,SB}}\left(\tfrac{1}{2}\right) &\overset{(1)}{=} 2\mathrm{Var}[Y_i(1)] + 2\mathrm{Var}[Y_i(0)] \\ &\quad - E\left[[E[Y_i(1)+Y_i(0)|S_i] - E[Y_i(1)+Y_i(0)]]^2\right] \\ &\overset{(2)}{=} \mathrm{Var}[E[Y_i(1)-Y_i(0)|S_i]] \\ &\quad + 2E[\mathrm{Var}[Y_i(1)|S_i] + \mathrm{Var}[Y_i(0)|S_i]], \end{aligned} \tag{16}$$

where (1) holds by (11) under $\tau(S_i) = 0$ and $\pi = 1/2$ and (2) by the Law of Total Variance and simple manipulations. From here, we can deduce that

$$\Sigma_{\mathrm{SS}}\left(\tfrac{1}{2}\right) \overset{(1)}{\leq} \Sigma_{\mathrm{BS,SB}}\left(\tfrac{1}{2}\right) \overset{(2)}{\leq} \Sigma_{\mathrm{BS}}\left(\tfrac{1}{2}\right), \tag{17}$$

where (1) holds by (16) and that $S_i = S(X_i)$, and (2) holds by (11) and the fact that CAR mechanism with strong balance (i.e., $\tau(s) = 0$ for all $s \in \mathcal{S}$) is the most efficient big strata CAR mechanism. In words, (17) indicates that the OLS estimator in an RCT with MPD is more efficient than the OLS estimator in an RCT with treatment assignment implemented with any big strata CAR method.

At this point, one may wonder if it is possible to find an alternative small strata mechanism that can be used to produce a more efficient ATE estimator. Under the assumption that MPD is implemented using the baseline covariate X_i, this turns out not to be possible. In particular, Bai et al. (2024a) show that $\Sigma_{\mathrm{SS}}\left(\frac{1}{2}\right)$ achieves the semiparametric efficiency bound for ATE estimator based on X_i when $\pi = 1/2$.

Under suitable conditions, Bai (2022) and Bai et al. (2023a) provide a consistent estimator of the asymptotic variance $\Sigma_{\mathrm{SS}}\left(\frac{1}{2}\right)$ in (15). We denote this estimator by $\hat{\Sigma}_{\mathrm{SS}}\left(\frac{1}{2}\right)$. By combining this estimator with (14), we can implement asymptotically valid inference for $H_0 : \theta_0 = \bar{\theta}$ vs. $H_1 : \theta_0 \neq \bar{\theta}$ for any $\bar{\theta} \in \mathbb{R}$ in the context of an RCT with treatment assignment implemented by MPD. Specifically, for any significance level $\alpha \in (0, 1)$, we propose to reject $H_0 : \theta_0 = \bar{\theta}$ according to the following rejection rule:

$$\phi_n = I\left[\left|\frac{\sqrt{n}(\hat{\theta}_{\mathrm{OLS}} - \bar{\theta})}{\hat{\Sigma}_{\mathrm{SS}}\left(\frac{1}{2}\right)}\right| > z_{1-\alpha/2}\right], \tag{18}$$

where $z_{1-\alpha/2}$ denotes the $(1-\alpha/2)$-quantile of $N(0, 1)$.

4 Regression adjustment

Section 3 has demonstrated that using CAR methods in RCT can generate efficiency gains in the ATE estimation. In particular, designing an RCT with CAR methods can reduce the asymptotic variance of $\hat{\theta}_{\text{OLS}}$. Recall that $\hat{\theta}_{\text{OLS}}$ is the result of the OLS regression in (5) without any additional covariates. This section explores an alternative way of achieving efficiency gains by covariate adjustment, i.e., by suitably adding covariates to the OLS regression in (5).

4.1 Big strata

This section revisits the "big strata" setup of Section 3.1. Given the baseline covariate data $\{X_i\}_{i=1}^n$, recall that the researcher defines strata according to the function $S : \mathcal{X} \to \mathcal{S} = \{1, \ldots, |\mathcal{S}|\}$ with $|\mathcal{S}| \in \mathbb{N}$, so that $S_i = S(X_i)$ and $p(s) \equiv P(S_i = s) > 0$ for all $s \in \mathcal{S}$. The main result in that section was that using CAR methods in the design stage can lead to weakly more efficient ATE inference by enforcing balanced assignments within each stratum. In the case of the CAR method with "strong balance" (i.e., $\tau(s) = 0$ for all $s \in \mathcal{S}$), the effect on the asymptotic variance of the OLS estimator in (4) is the same as if we had conditioned on S_i.

We show in this section that covariate adjustment in the analysis stage can be supplementary or complementary to using CAR methods in the design stage. In particular, Section 4.1.1 shows that the conditioning effects of CAR methods can be reproduced via regression adjustment based on strata indicators. In addition, Section 4.1.2 shows how to obtain further efficiency gains by regression adjustment with additional covariates.

4.1.1 Regression adjustment with strata indicators

To motivate the practice of regression adjustment, we begin by considering an ATE estimator that results from adding strata indicators to the linear regression in Bugni et al. (2019). Specifically, we consider the ATE estimator based on running the following "fully saturated" linear regression:

$$Y_i = \sum_{s \in \mathcal{S}} \delta_s I\{S_i = s\} + \sum_{s \in \mathcal{S}} \tau(s) A_i I\{S_i = s\} + \varepsilon_i \text{ for } i = 1, \ldots, n. \tag{19}$$

For all $s \in \mathcal{S}$, denote by $\hat{\delta}_s$ and $\hat{\tau}(s)$ the resulting OLS estimators of δ_s and $\tau(s)$, respectively. The corresponding estimator of the ATE is given by

$$\hat{\theta}_{\text{SAT}} = \sum_{s \in \mathcal{S}} \hat{p}(s) \hat{\tau}(s), \tag{20}$$

where $\hat{p}(s) = \frac{n(s)}{n}$ is an estimator of $p(s) = P(S_i = s)$ for any $s \in \mathcal{S}$.

Under Assumption 3.1, Bugni et al. (2019, Theorem 3.1) shows that

$$\sqrt{n}(\hat{\theta}_{\text{SAT}} - \theta_{\text{ATE}}) \xrightarrow{d} N(0, \Sigma_{\text{BS,SB}}(\pi)), \tag{21}$$

where $\Sigma_{\text{BS,SB}}(\pi)$ is the asymptotic variance of the "big strata" CAR method with "strong balance" (i.e., $\tau(s) = 0$ for all $s \in \mathcal{S}$), given in (12). It is important to note that (21) is obtained under Assumption 3.1 even if the treatment assignment mechanism does not use strong balance. For example, (21) applies even if treatment assignment is implemented via SRA (Example 2.2) or CRA (Example 2.3), which are both included as special cases of Assumption 3.1. In other words, one consequence of covariate adjustment via strata indicators in the manner described in (19) is that the ATE estimator attains the most efficient implementation of the CAR method. In this sense, we say that covariate adjustment in the analysis stage can *supplement* the use of CAR methods in the design stage. Regardless of whether CAR methods are used or not in the design stage, $\hat{\theta}_{\text{SAT}}$ in (20) is as efficient as $\hat{\theta}_{\text{OLS}}$ in (4) under the most efficient implementation of big strata CAR method allowed by Assumption 3.1.

To implement asymptotically valid inference based on (21), we only require a consistent estimator of $\Sigma_{\text{BS,SB}}(\pi)$. This can be obtained by specializing $\hat{\Sigma}_{\text{BS}}(\pi)$ mentioned in previous section to $\tau(s) = 0$ for all $s \in \mathcal{S}$; see Bugni et al. (2019) for details. This allows us to implement asymptotically valid inference for $H_0 : \theta_0 = \bar{\theta}$ vs. $H_1 : \theta_0 \neq \bar{\theta}$ using the rejection rule in (13) with $\hat{\theta}_{\text{SAT}}$ replaced by $\hat{\theta}_{\text{OLS}}$ and $\hat{\Sigma}_{\text{BS}}(\pi)$ replaced by $\hat{\Sigma}_{\text{BS,SB}}(\pi)$.

4.1.2 Regression adjustment with additional baseline covariates

The previous section considers the regression adjustment only using strata indicators. If additional covariates become available at the analysis stage, it is natural to use them to implement further regression adjustments. With some abuse of notation, let X_i denote the vector of original baseline variables used to construct the strata in the design stage (i.e., $S_i = S(X_i)$ in previous sections) plus additional covariates that are only available in the analysis stage. In the context of CAR with big strata, our goal is to use these covariates to produce a new ATE estimator that is guaranteed to be (weakly) more efficient than $\hat{\theta}_{\text{SAT}}$, and thus, $\hat{\theta}_{\text{OLS}}$.

Even in the context of RCTs without CAR, it is well known that adding a regressor X_i into a regression like (5) or (19) is not guaranteed to improve the efficiency of the resulting ATE estimator. That is, adding an arbitrary regressor into either of these regressions may result in an ATE estimator that is less efficient. This is known in the literature as the Freedman's critique; see Freedman (2008a,b). In RCTs without CAR, Lin (2013) shows that this critique can be overcome by adding covariates into these regressions in a principled manner. The rest of this section will review methods to address Freedman's critique in the context of RCTs with big strata CAR.

To frame the discussion for the regression adjustment, it is useful to consider an infeasible "ideal" regression-adjusted ATE estimator. For $a \in \{0, 1\}$, let $m_a^*(x, s) = E[Y_i(a)|X_i = x, S_i = s]$ denote the expectation of the potential outcome $Y_i(a)$ conditional on $X_i = x$ and $S_i = s$, and let $m_a^*(s) =$

$E[m_a^*(X_i, S_i)|S_i = s] = E[Y_i(a)|S_i = s]$.[4] With "oracle" knowledge of the functions $(m_1^*(x, s), m_0^*(x, s), m_1^*(x), m_0^*(x))$, one could compute the following adjusted outcome observations:

$$Y_i^{\text{ADJ},*} = Y_i - (m_{A_i}^*(X_i, S_i) - m_{A_i}^*(S_i)) \text{ for } i = 1, \ldots, n.$$

As it turns out, the saturated regression in (19) where the outcome data $\{Y_i\}_{i=1}^n$ is replaced by the adjusted outcome $\{Y_i^{\text{ADJ},*}\}_{i=1}^n$ would produce an estimator of the ATE that is semiparametrically efficient. Of course, the functions (m_1^*, m_0^*) are unknown objects to the researcher, so the regression mentioned above is unavailable in applications.

In practice, the researcher can propose a "working model" for (m_1^*, m_0^*), which we denote by (m_1, m_0). Also, let $(\hat{m}_1, \hat{m}_0)$ denote suitable estimators of the working model. Importantly, the working model is allowed to be misspecified and, consequently, $(\hat{m}_1, \hat{m}_0)$ need not consistently estimate the true conditional expectations in (m_1^*, m_0^*). We now illustrate with an example.

Example 4.1 (Linear model with strata indicators). A popular working model in applications is the linear model with strata indicators, i.e., for $i = 1, \ldots, n$,

$$m_1(x, s) = \sum_{\tilde{s} \in \mathcal{S}} 1\{\tilde{s} = s\} x' \beta_{1,\tilde{s}} \quad \text{and} \quad m_0(x, s) = \sum_{\tilde{s} \in \mathcal{S}} 1\{\tilde{s} = s\} x' \beta_{0,\tilde{s}}, \tag{22}$$

where $\{(\beta_{1,s}, \beta_{0,s}) \in \mathbb{R}^{2d_X} : s \in \mathcal{S}\}$ are model coefficients to be estimated. By replacing these unknown coefficients by suitable estimators $\{(\hat{\beta}_{1,s}, \hat{\beta}_{0,s}) \in \mathbb{R}^{2d_X} : s \in \mathcal{S}\}$, we obtain the estimated working model:

$$\hat{m}_1(x, s) = \sum_{\tilde{s} \in \mathcal{S}} 1\{\tilde{s} = s\} x' \hat{\beta}_{1,\tilde{s}} \quad \text{and} \quad \hat{m}_0(x, s) = \sum_{\tilde{s} \in \mathcal{S}} 1\{\tilde{s} = s\} x' \hat{\beta}_{0,\tilde{s}}. \tag{23}$$

As mentioned earlier, our results will not require the working model in (22) to be a correctly specified model for (m_1^*, m_0^*). □

We can evaluate the working model for each observation $\{\hat{m}_{A_i}(X_i, S_i)\}_{i=1}^n$, where

$$\hat{m}_{A_i}(X_i, S_i) = \sum_{a \in \{0,1\}} \sum_{\tilde{s} \in \mathcal{S}} 1\{S_i = s, A_i = a\} \hat{m}_a(x, s)$$

Furthermore, we can also compute $\{\hat{m}_{A_i}(S_i)\}_{i=1}^n$, where

$$\hat{m}_a(s) = \frac{\sum_{j=1}^n 1\{S_j = s, A_j = a\} \hat{m}_{A_j}(X_j, S_j)}{\sum_{j=1}^n 1\{S_j = s, A_j = a\}}.$$

[4] Note that when S_i is a function of X_i, then conditioning on S_i becomes redundant, i.e., $E[Y_i(a)|X_i = x, S_i = s] = E[Y_i(a)|X_i = x]$.

Given these estimators, we can compute the feasible adjusted outcome data:

$$Y_i^{\text{ADJ}} = Y_i - (\hat{m}_{A_i}(X_i, S_i) - \hat{m}_{A_i}(S_i)) \text{ for } i = 1, \ldots, n.$$

We can then use the adjusted outcome data to run an SAT regression in (19), i.e.,

$$Y_i^{\text{ADJ}} = \sum \delta_s^{\text{ADJ}} I\{S_i = s\} + \sum \tau^{\text{ADJ}}(s) A_i I\{S_i = s\} + \varepsilon_i \text{ for } i = 1, \ldots, n.$$

For all $s \in \mathcal{S}$, denote by $\hat{\delta}_s^{\text{ADJ}}$ and $\hat{\tau}^{\text{ADJ}}(s)$ the resulting OLS estimators of δ_s^{ADJ} and $\tau^{\text{ADJ}}(s)$, respectively. The corresponding regression-adjusted estimator of the ATE is given by

$$\hat{\theta}_{\text{ADJ}} = \sum_{s \in \mathcal{S}} \hat{p}(s) \hat{\tau}^{\text{ADJ}}(s), \tag{24}$$

where $\hat{p}(s) = n(s)/n$ is the estimator of $p(s) = P(S_i = s)$ for any $s \in \mathcal{S}$.

Regardless of the specification of the working model, the proposed regression-adjusted ATE estimator is consistent and asymptotically normal. To gain intuition of this result, it is useful to note that $\hat{\theta}_{\text{ADJ}}$ is, in essence, an augmented inverse probability weighted (AIPW) estimator of the ATE (Robins et al., 1994). To see this, we note that $\hat{\theta}_{\text{ADJ}}$ in (24) can be reexpressed in the following manner:

$$\begin{aligned}
\hat{\theta}_{\text{ADJ}} &= \left\{ \begin{array}{c} \frac{1}{n}\sum_{i=1}^n \frac{A_i(Y_i - \hat{m}_1(X_i, S_i))}{\hat{\pi}(S_i)} - \frac{1}{n}\sum_{i=1}^n \frac{(1-A_i)(Y_i - \hat{m}_0(X_i, S_i))}{1-\hat{\pi}(S_i)} \\ + \frac{1}{n}\sum_{i=1}^n (\hat{m}_1(X_i, S_i) - \hat{m}_0(X_i, S_i)) \end{array} \right\} \\
&= \frac{1}{n}\sum_{i=1}^n \frac{A_i(Y_i - (\hat{m}_1(X_i, S_i) - \hat{m}_1(S_i)))}{\hat{\pi}(S_i)} \\
&\quad - \frac{1}{n}\sum_{i=1}^n \frac{(1 - A_i)(Y_i - (\hat{m}_0(X_i, S_i) - \hat{m}_0(S_i)))}{1 - \hat{\pi}(S_i)},
\end{aligned}$$

where $\hat{\pi}(s) = n_1(s)/n(s)$ for any $s \in \mathcal{S}$. In this AIPW formulation, $\hat{\pi}(s)$ and $(\hat{m}_0, \hat{m}_1))$ are the estimators for the propensity score π and conditional mean function (m_1^*, m_0^*), respectively. Because the AIPW moment is "doubly robust" with respect to these two nuisance parameters (i.e., propensity score and the conditional means), and $\hat{\pi}(s)$ is a consistent estimator of π, it follows that $\hat{\theta}_{\text{ADJ}}$ is a consistent and asymptotically normal estimator of the ATE, regardless of whether $(\hat{m}_0, \hat{m}_1)$ consistently estimates the true conditional expectations (m_1^*, m_0^*).

For the remainder of this section, we focus on the regression adjustment using the linear regression working model in Example 4.1. Different choices of coefficients $\{(\beta_{0,s}, \beta_{1,s}) : s \in \mathcal{S}\}$ in (22) generate distinct regression-adjusted estimators $\hat{\theta}_{\text{ADJ}}$. For instance, setting $\beta_{0,s} = \beta_{1,s} = 0$ for all $s \in \mathcal{S}$ yields the estimator $\hat{\theta}_{\text{SAT}}$ discussed in Section 4.1.1 with corresponding asymptotic variance

$\Sigma_{BS,SB}(\pi)$. It is then natural to search for the optimal set of linear coefficients $\{(\beta^*_{0,s}, \beta^*_{1,s}) : s \in \mathcal{S}\}$, defined as those that minimize the asymptotic variance of $\hat{\theta}_{\text{ADJ}}$ across all possible coefficients. It can be shown that these optimal coefficients solve the following equation:

$$\begin{aligned}&(1-\pi)\beta^*_{1,s} + \pi\beta^*_{0,s}\\ &\quad = \text{Var}(X_i|S_i = s)^{-1}\,\text{Cov}(X_i, (1-\pi)Y_i(1) + \pi Y_i(0)|S_i = s) \text{ for all } s \in \mathcal{S}.\end{aligned} \tag{25}$$

There are infinitely many solutions to (25). Among these solutions, it is natural to pick the following one: for all $s \in \mathcal{S}$,

$$\begin{aligned}\beta^*_{1,s} &= \text{Var}(X_i|S_i = s)^{-1}\text{Cov}(X_i, Y_i(1)|S_i = s),\\ \beta^*_{0,s} &= \text{Var}(X_i|S_i = s)^{-1}\text{Cov}(X_i, Y_i(0)|S_i = s).\end{aligned} \tag{26}$$

With some abuse of notation, we refer to these as *the* optimal linear coefficients.

We can obtain a consistent estimator for the linear coefficients $\{(\beta^*_{0,s}, \beta^*_{1,s}) : s \in \mathcal{S}\}$ by running OLS regressions with different subsets of the data. For each $(a, s) \in \{0, 1\} \times \mathcal{S}$, we use the subsample of observations with $S_i = s$ and $A_i = a$ to run the following OLS regression:

$$Y_i = \theta^*_{a,s} + (X_i - \bar{X}(s))'\beta^*_{a,s} + u_i, \tag{27}$$

where

$$\bar{X}(s) = \frac{\sum_{i=1}^n 1\{S_i = s\}X_i}{\sum_{j=1}^n 1\{S_j = s\}}.$$

This regression yields the following OLS estimator

$$\begin{aligned}\hat{\beta}^*_{a,s} = &\left(\sum_{i=1}^n (X_i - \bar{X}_{a,s})(X_i - \bar{X}_{a,s})' I\{S_i = s, A_i = a\}\right)^{-1}\\ &\times \left(\sum_{i=1}^n (X_i - \bar{X}_{a,s})Y_i I\{S_i = s, A_i = a\}\right),\end{aligned}$$

where

$$\bar{X}_{a,s} = \frac{\sum_{i=1}^n 1\{S_i = s, A_i = a\}X_i}{\sum_{j=1}^n 1\{S_j = s, A_j = a\}}.$$

By plugging these results on (24), the regression-adjusted estimator with the optimal linear coefficient is denoted as $\hat{\theta}^*_{\text{ADJ}}$ and defined as

$$\hat{\theta}^*_{\text{ADJ}} = \sum_{s \in \mathcal{S}} \hat{p}(s)\hat{\tau}^*(s), \tag{28}$$

where $\hat{\tau}^*(s)$ is the OLS estimator of $\tau^*(s)$ in the regression

$$Y_i = \sum_{s \in \mathcal{S}} \left(\begin{array}{c} \alpha_s^* I\{S_i = s\} + \tau^*(s) I\{S_i = s\} A_i \\ +(X_i - \bar{X}(s))' \mu_s^* I\{S_i = s\} + (X_i - \bar{X}(s))' \delta_s^* I\{S_i = s\} A_i \end{array} \right) + \varepsilon_i.$$

This is precisely the estimator proposed by Ye et al. (2022).

Under Assumption 3.1 and other suitable conditions, it is possible to show that

$$\sqrt{n}(\hat{\theta}^*_{\text{ADJ}} - \theta_{\text{ATE}}) \xrightarrow{d} N(0, \Sigma_{\text{BS,ADJ}}(\pi)), \tag{29}$$

where $\Sigma_{\text{BS,ADJ}}(\pi)$ is the asymptotic variance of the "big strata" CAR method with "strong balance" (i.e., $\tau(s) = 0$ for all $s \in \mathcal{S}$) and regression adjustment according to the working model $(\hat{m}_1, \hat{m}_0)$. Moreover, under mild conditions regarding the working model, it is possible to establish that

$$\Sigma_{\text{BS,ADJ}}(\pi) \leq \Sigma_{\text{BS,SB}}(\pi). \tag{30}$$

In words, the regression-adjusted ATE estimator $\hat{\theta}_{\text{ADJ}}$ is guaranteed to be as efficient as the non-adjusted ATE estimators $\hat{\theta}_{\text{SAT}}$ or $\hat{\theta}_{\text{OLS}}$. This implies that the proposed ATE estimator $\hat{\theta}_{\text{ADJ}}$ addresses the Freedman's critique.

4.2 Small strata

As in Section 3.2, this section considers an RCT implemented with matched pair design (MPD) based on the covariates $\{X_i\}_{i=1}^n$. We have shown in that section that the MPD exhausts all the information available in $\{X_i\}_{i=1}^n$. Thus, if no other covariate information is available, regression adjustment becomes unnecessary. To make the regression adjustment problem relevant, we assume that the researcher has additional covariates $\{W_i\}_{i=1}^n$ (where W_i is not a function of X_i) that can be used to conduct regression adjustment in the analysis stage. This section explores how to implement this regression adjustment.

As in the previous section, the "ideal" regression-adjusted ATE estimator would be based on expectations of the potential outcome conditional on all covariates, i.e., $m_a^*(x, w) = E[Y_i(a)|X_i = x, W_i = w]$ for $a \in \{0, 1\}$. Of course, this conditional expectation is unknown in practice. As in the previous section, we can build the regression-adjusted estimator based on the AIPW formulation (Robins et al., 1994). For a "working model" $(m_1(x, w), m_0(x, w))$, the AIPW moment conditions are

$$\begin{aligned} E[Y_i(1)] &= E[2A_i(Y_i - m_1(X_i, W_i)) + m_1(X_i, W_i)], \\ E[Y_i(0)] &= E[2(1 - A_i)(Y_i - m_0(X_i, W_i)) + m_0(X_i, W_i)], \end{aligned} \tag{31}$$

where the treatment probability is set to $\pi = 1/2$. We note that this treatment probability is correctly specified in the context of an MPD. By this and the

double robustness property, the method is robust to the misspecification of the working model (m_1, m_0). Based on the moment condition in (31), the regression adjusted estimator of θ_{ATE} is given by

$$\hat{\theta} = \hat{\mu}(1) - \hat{\mu}(0), \tag{32}$$

where, for $a \in \{0, 1\}$,

$$\hat{\mu}(a) = \frac{1}{n}\sum_{i=1}^{n}(2I\{A_i = a\}(Y_i - \hat{m}_a(X_i, W_i)) + \hat{m}_a(X_i, W_i)) \tag{33}$$

and $(\hat{m}_1, \hat{m}_0)$ is a suitable estimator of the working model (m_1, m_0). This formulation of the regression adjusted estimator for ATE under MPDs was independently discovered by Bai et al. (2023b) and Cytrynbaum (2023).

As in Section 4.1.2, it is relevant to focus on the class of linear working models, i.e.,

$$m_0(X_i, W_i) = \psi_i'\beta_0 \text{ and } m_1(X_i, W_i) = \psi_i'\beta_1, \tag{34}$$

where $\psi_i = \psi(X_i, W_i)$ is a function of (X_i, W_i) that satisfies suitable regularity conditions.[5] Furthermore, let $(\hat{\beta}_0, \hat{\beta}_1)$ denote consistent estimators of (β_0, β_1). By plugging these on (33), we obtain the following estimated working model:

$$\hat{m}_0(X_i, W_i) = \psi_i'\hat{\beta}_0 \text{ and } \hat{m}_1(X_i, W_i) = \psi_i'\hat{\beta}_1.$$

By using these on (32), we derive the following linear-adjusted ATE estimator:

$$\begin{aligned}\hat{\theta}_{\text{ADJ}} = &\frac{1}{n}\sum_{i=1}^{n}(2A_i(Y_i - \psi_i'\hat{\beta}_1) + \psi_i'\hat{\beta}_1)\\ &- \frac{1}{n}\sum_{i=1}^{n}(2(1 - A_i)(Y_i - \psi_i'\hat{\beta}_0) + \psi_i'\hat{\beta}_0).\end{aligned} \tag{35}$$

If we further impose $\hat{\beta}_1 = \hat{\beta}_0 = \hat{\beta}$, $\hat{\theta}_{\text{ADJ}}$ simplifies to

$$\hat{\theta}_{\text{ADJ}} = \frac{2}{n}\sum_{i=1}^{n}(2A_i - 1)(Y_i - \psi_i'\hat{\beta}).$$

As in Section 4.1.2, the adjusted estimator $\hat{\theta}_{\text{ADJ}}$ is consistent and asymptotically normal, and its asymptotic variance depends on the choice of (β_0, β_1). For instance, setting $\hat{\beta}_1 = \hat{\beta}_0 = \hat{\beta} = 0$, $\hat{\theta}_{\text{ADJ}}$ reduces to the OLS estimator $\hat{\theta}_{\text{OLS}}$ studied in Section 3.2. In the rest of this section, we aim to search for the optimal linear coefficient so that the corresponding regression-adjusted estimator

[5] Specifically, we require no component of ψ is constant and $E[\text{Var}[\psi_i|X_i]]$ is non-singular.

can achieve the smallest asymptotic variance among all linearly adjusted ATE estimators. By construction, the ATE estimator with the optimal linear coefficients is weakly more efficient than the OLS estimator $\hat{\theta}_{\text{OLS}}$, addressing the Freedman's critique.

It can be shown that the optimal (β_0, β_1), denoted as (β_0^*, β_1^*), satisfies the following equation

$$\beta_1^* + \beta_0^* = (E[\text{Var}[\psi_i|X_i]])^{-1} E[\text{Cov}[\psi_i, Y_i(1) + Y_i(0)|X_i]].$$

The optimal linear coefficients are not uniquely defined. For simplicity, we choose the solution given by $\beta_0^* = \beta_1^* = \beta^*$ with

$$\beta^* = (2E[\text{Var}[\psi_i|X_i]])^{-1} E[\text{Cov}[\psi_i, Y_i(1) + Y_i(0)|X_i]].$$

The next goal is to find a consistent estimator for β^*, or, alternatively, to derive a linear regression representation of $\hat{\theta}_{\text{ADJ}}$ that implements the optimal linear regression adjustment.

By trial and error, we first consider estimating by OLS the following linear regression

$$Y_i = \alpha + \theta A_i + \psi_i' \beta + \epsilon_i, \tag{36}$$

and denote the estimator of θ as $\hat{\theta}_{\text{naive}}$. It follows from a direct calculation that

$$\hat{\theta}_{\text{naive}} = \frac{2}{n} \sum_{i=1}^{n} (Y_i - \psi_i' \hat{\beta}^{\text{naive}})(2A_i - 1).$$

Therefore, $\hat{\theta}_{\text{naive}}$ satisfies (32)–(33) with

$$\hat{m}_a(X_i, W_i) = \psi_i' \hat{\beta}^{\text{naive}},$$

which means $\hat{\theta}_{\text{naive}}$ is indeed a linear regression adjusted estimator with linear coefficient $\hat{\beta}_1 = \hat{\beta}_0 = \hat{\beta}^{\text{naive}}$. However, Bai et al. (2023b) show that

$$\hat{\beta}^{\text{naive}} \xrightarrow{p} \beta^{\text{naive}} = \text{Var}[\psi_i]^{-1} \text{Cov}[\psi_i, Y_i(1) + Y_i(0)],$$

and $\beta^{\text{naive}} \neq \beta^*$, which implies $\hat{\theta}_{\text{naive}}$ is, in general, not the optimally linearly adjusted estimator.

The key difference between β^{naive} and β^* lies in the conditional variance and covariance calculations in the latter, which are taken conditionally on the matching variable X_i. In principle, one could consider estimating these conditional variances and covariances by using standard nonparametric methods. However, in the context of MPDs, Bai et al. (2023b) show that one can alternatively achieve this goal by running linear regressions with strata fixed effects. The basic insight is that each pair of observations formed by MPD necessarily shares a similar value of X_i. Thus, introducing the pair dummies effectively

amounts to "controlling" for X_i. Specifically, Bai et al. (2023b) propose the following linear regression with $[n/2]$ strata fixed effects:

$$Y_i = \theta A_i + \psi_i' \beta + \sum_{s=1}^{[n/2]} \gamma_s I\{S_i = s\} + \epsilon_i. \tag{37}$$

We denote the OLS estimators of θ and β from this regression $\hat{\theta}_{\text{PFE}}$ and $\hat{\beta}_{\text{PFE}}$. Indeed, we have

$$\hat{\theta}_{\text{PFE}} = \frac{2}{n} \sum_{i=1}^{n} (Y_i - \psi_i' \hat{\beta}_{\text{PFE}})(2A_i - 1).$$

Bai et al. (2023b) further show that

$$\hat{\beta}_{\text{PFE}} \xrightarrow{p} \beta^*,$$

which implies

$$\sqrt{n}(\hat{\theta}_{\text{PFE}} - \theta_{\text{ATE}}) \xrightarrow{d} N(0, \Sigma_{\text{SS,ADJ}}(1/2))$$

with

$$\Sigma_{\text{SS,ADJ}}(1/2) \leq \Sigma_{\text{SS}}(1/2).$$

In addition, we have $\Sigma_{\text{SS,ADJ}}(1/2) = \Sigma_{\text{SS}}(1/2)$ only when W_i is a function of X_i and $\beta^* = 0$. Bai et al. (2023b) provide the exact expression and consistent estimator for $\Sigma_{\text{SS,ADJ}}(1/2)$.

5 Imperfect compliance

In this section, we allow for the possibility that RCT participants fail to comply with the assigned treatment. As in previous sections, assume that the researcher selects a random sample of size n to implement the RCT. For each $i = 1, \ldots, n$, we use $A_i \in \{0, 1\}$ to indicate the treatment assigned to the individual i, Y_i to denote the outcome of individual i, and X_i to denote the vector of baseline covariates of individual i. In addition, we introduce $D_i \in \{0, 1\}$ to indicate the actual treatment decision adopted by individual i. Imperfect compliance implies that the treatment decision may not coincide with the treatment assignment, i.e., $D_i \neq A_i$ for some $i = 1, \ldots, n$.

We model treatment decisions via a potential decision model, where $D_i(A)$ denotes the potential treatment decision of individual i under treatment assignment A. Following the usual classification in Angrist and Imbens (1994), each RCT participant can only be one of four types: complier, always taker, never taker, or defier. An individual i is said to be a complier if $\{D_i(0) = 0, D_i(1) =$

1}, an always taker if $\{D_i(0) = D_i(1) = 1\}$, a never taker if $\{D_i(0) = D_i(1) = 0\}$, and a defier if $\{D_i(0) = 1, D_i(1) = 0\}$. For the potential outcomes model, we assume that outcomes are a function of the treatment decision but not the treatment assignment. That is, we use $Y_i(D)$ to denote the potential outcome of individual i under treatment decision D (and note that the treatment assignment A is excluded). These potential decision and outcome models are related to their observed counterparts in the usual manner:

$$\begin{aligned} D_i &= D_i(1)A_i + D_i(0)(1 - A_i), \\ Y_i &= Y_i(1)D_i + Y_i(0)(1 - D_i). \end{aligned} \tag{38}$$

In this context, we allow the researcher to implement the RCT with CAR. For simplicity and concreteness, we restrict attention to the big strata CAR framework described in Section 3.1.[6] That is, given baseline covariate data $\{X_i\}_{i=1}^n$ with support $\mathcal{X}$, the researcher defines strata according to a pre-specified function $S : \mathcal{X} \to \mathcal{S} = \{1, \ldots, |\mathcal{S}|\}$ with $|\mathcal{S}| \in \mathbb{N}$. According to this function, individual i is assigned to strata $S_i = S(X_i)$. We again rule out irrelevant strata by assuming $p(s) = P(S_i = s) > 0$ for all $s \in \mathcal{S}$.

Example 5.1 (Running example, continued). In the hypothetical RCT in Example 2.1, recall that $A_i \in \{0, 1\}$ indicates whether individual i is assigned to the training or not. In this context, imperfect compliance would occur either when individuals assigned to the training may decide not to attend their training session or when individuals assigned to control find an alternative way of receiving their training. That is, we can have $D_i \neq A_i$, where $D_i \in \{0, 1\}$ indicates whether individual i actually underwent the training or not.

By definition, a complier is an individual who gets training if and only if assigned to it, an always taker is an individual who decides to get training regardless of assignment, a never taker is an individual who decides not to get training regardless of assignment, and a defier is an individual who gets training if and only if *not* assigned to it. The defier seems to be an unusual type, and this helps motivate the assumption that there are no defiers in the population of interest. □

A key complication with imperfect compliance is that the ATE in (2) is no longer identified. To explore this, consider the following decomposition based on the law of iterated expectations:

$$\begin{aligned} \theta_{\text{ATE}} &= E[Y_i(1) - Y_i(0)] \\ &= \left\{ \begin{array}{l} E[Y_i(1) - Y_i(0) | D_i(0) = 0, D_i(1) = 1] P(D_i(0) = 0, D_i(1) = 1) \\ + E[Y_i(1) - Y_i(0) | D_i(0) = D_i(1) = 1] P(D_i(0) = D_i(1) = 1) \\ + E[Y_i(1) - Y_i(0) | D_i(0) = D_i(1) = 0] P(D_i(0) = D_i(1) = 0) \\ + E[Y_i(1) - Y_i(0) | D_i(0) = 1, D_i(1) = 0] P(D_i(0) = 1, D_i(1) = 0) \end{array} \right\}. \end{aligned}$$

[6] To our knowledge, the results covered in this section for the small-strata framework (e.g., matched pairs) have not yet been investigated.

The equation decomposes the ATE into the four possible individual types: compliers, always takers, never takers, and defiers. To understand the identification issue, note that individuals who are either never or always takers do not get to reveal one of their potential outcomes, regardless of their treatment assignment. Never takers never reveal their outcome under treatment (i.e., $Y_i(1)$), and always takers never reveal their outcome under control (i.e., $Y_i(0)$). We note that this identification issue arises regardless of whether one assumes that defiers are present in the data or not.

In light of this issue, one natural reaction is to pursue alternative estimands that are of practical relevance and point identified under imperfect compliance. To illustrate, consider the intention to treat (ITT) parameter, which corresponds to the average outcome of being assigned to treatment vs. being assigned to control. Within our potential outcomes and decisions framework, the ITT estimand is given by

$$\theta_{\text{ITT}} \equiv E\left[Y_i | A_i = 1\right] - E\left[Y_i | A_i = 0\right].$$

While the ITT estimand is point identified, it does not have a natural causal interpretation in our current setup.[7]

The remainder of this section describes two approaches to inference under imperfect compliance. Section 5.1 considers inference on the local average treatment effect (LATE), which is a new estimand that has a causal interpretation and is point identified under suitable conditions. Finally, Section 5.2 revisits our original estimand of interest, the ATE, using partially identified inference methods.

5.1 Inference on the local average treatment effect (LATE)

One estimand that has a clear causal interpretation is the so-called local average treatment effect (LATE), which is defined as the average treatment effect for the sub-population of compliers, i.e.,

$$\theta_{\text{LATE}} \equiv E\left[Y_i(1) - Y_i(0) | D_i(0) = 0, D_i(1) = 1\right].$$

The parameter was first proposed in the seminar contribution by Imbens and Angrist (1994). This paper shows that the LATE is identified under the no-defiers assumption, which indicates that the population of interest has no defiers: $P\left(D_i(0) = 1, D_i(1) = 0\right) = 0$.[8] As discussed in Example, 5.1, the no-defiers assumption is a reasonable condition in many empirical settings. In other words, it

[7] Under our CAR framework and in the absence of compliers, one can show that θ_{ITT} equals the weighted average of treatment effect conditional on strata and being a compiler, multiplied by the probability of compiler conditional on the strata. Thus, the ITT is not an average of treatment effects even in the favorable case of treatment effect homogeneity.

[8] The no-defiers assumption is also known as the monotonicity assumption, as it imposes that the treatment decision is a weakly monotonically increasing function of the treatment assignment: $P\left(D_i(0) \leq D_i(1)\right) = 1$.

requires that there are no agents who always choose the exact opposite of their assigned treatment. Under the no-defiers assumption, the LATE is identified, as well as the probabilities of compliers, always takers, and never takers.

Bugni and Gao (2023) provide consistent and asymptotically normal estimators of the LATE under a CAR framework. These estimators are shown to be the result of instrumental variable (IV) regressions of the outcome variable on the treatment decisions and strata indicators, where the (endogenous) treatment decisions are instrumented by the (conditionally exogenous) treatment assignments. The simplest possible LATE estimator is given by

$$\hat{\theta}_{\mathrm{IV}} = \frac{\widehat{\mathrm{Cov}}\,(A, Y)}{\widehat{\mathrm{Cov}}\,(A, D)}. \tag{39}$$

By definition, $\hat{\theta}_{\mathrm{IV}}$ is the IV analog of (4), as it is β estimator result of the following IV regression

$$Y_i = \alpha + \beta D_i + \varepsilon_i \quad \text{for } i = 1, \ldots, n,$$

where D_i is instrumented by A_i.

Under the CAR assumptions and other suitable conditions, Bugni and Gao (2023) show that

$$\sqrt{n}(\hat{\theta}_{\mathrm{IV}} - \theta_{\mathrm{LATE}}) \xrightarrow{d} N\,(0, \Sigma_{\mathrm{IV}})\,, \tag{40}$$

where Σ_{IV} is a function of parameters of the DGP and the treatment assignment scheme. As in Section 3.1, Σ_{IV} is a function of the balancing parameter $\{\tau\,(s) : s \in S\}$ in Assumption 3.1, and it is minimized when the CAR mechanism is strongly balancing, i.e., $\tau\,(s) = 0$ for all $s \in S$. The paper provides a consistent estimator of the asymptotic variance in (40); see Bugni and Gao (2023, Theorems 5.1-5.2) for details. Based on this, we can implement asymptotically valid inference for $H_0 : \theta_{\mathrm{LATE}} = \bar{\theta}$ vs. $H_1 : \theta_{\mathrm{LATE}} \neq \bar{\theta}$ for any $\bar{\theta} \in \mathbb{R}$. In particular, for any significance level $\alpha \in (0, 1)$, we propose to reject $H_0 : \theta_{\mathrm{LATE}} = \bar{\theta}$ according to the following rejection rule:

$$\phi_n = I\left[\frac{\sqrt{n}|\hat{\theta}_{\mathrm{IV}} - \bar{\theta}|}{\hat{\Sigma}_{\mathrm{IV}}^{1/2}} > z_{1-\alpha/2}\right], \tag{41}$$

where $\hat{\Sigma}_{\mathrm{IV}}$ is a consistent estimator for Σ_{IV} and $z_{1-\alpha/2}$ denotes the $(1-\alpha/2)$-quantile of $N(0, 1)$.

Besides (39), Bugni and Gao (2023) consider two other LATE estimators constructed by running other IV regression of Y_i on linear functions of D_i and strata indicators, where D_i is instrumented by A_i. These IV regressions can give rise to SAT LATE estimator, which is the regression-adjusted version of (39). The paper also characterizes the asymptotic distribution of these estimators and provides consistent estimators of their asymptotic variance to implement inference. Finally, it is worth noting that one could include additional adjustment terms to the IV regressions based on the baseline covariate information along the lines of Section 4.

5.2 Inference on the ATE

The previous section studied inference for the LATE parameter in a general CAR framework. As mentioned, the LATE has the causal interpretation of being the ATE for individuals among compliers. Unfortunately, in the context of treatment effect heterogeneity, the conclusions about the LATE do not extend beyond the complier group. Moreover, the analysis in the previous section requires the no-defiers assumption, which may fail to hold practice. To deal with this criticism, this section describes recent advances to conduct inference for the ATE (and other arguably policy-relevant estimands) based on partially identified inference techniques.

As explained earlier, the ATE cannot be uniquely determined under imperfect compliance. In this setting, it is natural to consider inference for the identified set of the ATE, i.e., the smallest set of real-valued parameters that include this estimand. There are many references that derive the identified set of the ATE under imperfect compliance when the data are i.i.d. (e.g., Manski, 1989, 1990, 1994; Balke and Pearl, 1997; Heckman and Vytlacil, 2001; Kitagawa, 2021). Under CAR, however, the data are not i.i.d., so the results from this literature do not apply.

Identification and inference for the ATE under CAR framework with imperfect compliance are investigated in a recent paper by Bugni et al. (2024).[9] Specifically, Bugni et al. (2024, Theorem 3.1) derives that the identified set of θ_{ATE} is $\Theta_I = [\theta^L_{\text{ATE}}, \theta^H_{\text{ATE}}]$, given by

$$\theta^L_{\text{ATE}} = E\left[\frac{(Y_i D_i + Y_L(1-D_i))}{P(A_i = 1|S^{(n)})} - \frac{(Y_i D_i + Y_H(1-D_i))}{P(A_i = 0|S^{(n)})}\right],$$
$$\theta^H_{\text{ATE}} = E\left[\frac{(Y_i D_i + Y_H(1-D_i))}{P(A_i = 1|S^{(n)})} - \frac{(Y_i D_i + Y_L(1-D_i))}{P(A_i = 0|S^{(n)})}\right],$$

where Y_L and Y_H are the logical lower and upper bounds for the potential outcomes. We note that these bounds apply even if the potential outcomes have no logical bounds (i.e., $Y_L = -\infty$ or $Y_H = \infty$) and, in that case, θ_{ATE} is completely unrestricted (i.e., $\theta^L_{\text{ATE}} = -\infty$ and $\theta^H_{\text{ATE}} = \infty$).

In a recent paper, Bugni et al. (2024) study inference methods for the ATE in the context of an RCT implemented with CAR. Specifically, this paper provides a consistent estimator of the identified set of θ_{ATE} and an asymptotically valid confidence interval that covers θ_{ATE} with a minimum desired coverage probability. The estimator of the identified set $\Theta_I = [\theta^L_{\text{ATE}}, \theta^H_{\text{ATE}}]$ is given by

$$\hat{\Theta}_I = [\hat{\theta}^L_{\text{ATE}}, \hat{\theta}^H_{\text{ATE}}],$$

[9] We focus our description on results for the ATE, but the paper also covers the average treatment on the treated (ATT) or the average treatment effect on the untreated (ATU).

where $\hat{\theta}^L_{\text{ATE}}$ and $\hat{\theta}^H_{\text{ATE}}$ are consistent estimators of θ^L_{ATE} and θ^H_{ATE}, respectively. The confidence interval is given by

$$\hat{C}_{1-\alpha} = [\hat{\theta}^L_{\text{ATE}} - \hat{c}^L, \theta^H_{\text{ATE}} + \hat{c}^H],$$

where $\hat{c}^L$ and $\hat{c}^H$ are suitably chosen to guarantee that θ_{ATE} belongs to $\hat{C}_{1-\alpha}$ with probability $(1-\alpha)$ or more as the sample size diverges. We defer to this paper for details. It is relevant to highlight that these results are derived within the CAR framework, so they do not follow from existing results in the literature of inference for the ATE under i.i.d. assumptions.

6 Conclusions

This paper highlights recent econometric developments in the analysis of randomized controlled trials (RCTs), focusing on how baseline covariates can improve the quality of inference both at the design and analysis stages. Covariate adaptive randomization and regression adjustment are two key methods that enhance the accuracy of treatment effect estimators in RCTs. Moreover, the analysis extends to RCTs with imperfect compliance, a common challenge that complicates both the identification and inference of the average treatment effect (ATE). By addressing these econometric techniques and issues, the paper provides valuable insights into improving the precision and reliability of causal estimates derived from RCTs, which continue to be a vital tool across various fields of research in economics.

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